A-A-51684A
21 December 1987
SUPERSEDING
A-A-51684
29 January 1987

COMMERCIAL ITEM DESCRIPTION

CONTROLLER, DROP RATE, INTRAVENOUS INJECTION

The General Services Administration has authorized the use of this commercial item description.

This commercial item description covers a disposable controller, drop rate, intravenous injection, for use with 20 gage or larger hypodermic needles.

Salient characteristics:

General. The fluid path of the intravenous injection controller shall be sterile and non-pyrogenic.

Characteristics. The flow regulator shall be suitable for use in the administration of common parenteral fluids. The intravenous injection controller shall have a female Luer lock on the inlet and a male Luer slip on the outlet.

A flow regulator shall be provided to control the fluid flow between 5 and 250 ml/hr. As a minimum, the regulator shall have 10 different graduations in multiples of mls/hr.

The plastic components coming in contact with the fluid shall meet the requirements of the Pharmacopeia of the United States for Class II plastics (USP). Those rubber and plastic components coming in contact with the fluid and end protectors shall contain not more than 5 parts per million total of lead, barium and cadmium.

The tubing shall have an overall length of 18" ± 1-1/2". The tubing shall be permanently attached in a leakproof manner to the flow regulator. The plastic tubing shall be clear, colorless, flexible polyvinyl chloride with a hardness of 65 ± 10 Shore A durometer when tested in accordance with ASTM D2240.

AMSC N/A FSC 6515

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

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A "Y" type injection site fabricated from clear or translucent rigid plastic shall be connected to the tubing below the flow regulator. Two of the legs shall be inline and the third leg shall be at a 30 to 36 degree angle to the tubing. The top leg of the inline portion of the "Y" shall have a resealable diaphragm or plug. Tubing shall be attached to the other two legs. The diaphragm or plug shall be molded from natural rubber. The diaphragm or plug shall be so secured that after puncture and withdrawal of the needle, there shall be no separation or leakage.

The inlet and outlet connectors shall be fabricated from rigid plastic containing no exudable toxic materials. An end protector shall be attached to each end connector. Each end protector shall be readily removable. When placed over each open end of the intravenous injection controller, it shall, after sterilization, maintain sterility and integrity of the entire fluid path of the device until manually removed.

Luer fittings. Both the male and female Luer fittings shall meet the dimensional and performance requirements of ANSI/HIMA MD70.1.

Tests.

Freedom from pyrogens. The fluid path of the intravenous injection controller shall be free from pyrogens when tested in accordance with the USP.

Sterility test. The intravenous injection sets shall be tested for sterility in accordance with the method specified in the USP using inoculated product, inoculated simulated product, or paper spore strip (carrier). Product containing a biological indicator shall be marked or labeled so that it clearly differentiates it from product intended for distribution. The incubation medium for the paper spore strip (carrier) shall be soybean-casein digest. The incubation media for the inoculated product, or inoculated simulated product shall be fluid thioglycollate or alternative thioglycollate at 30° to 35°C for one half of the specimens and soybean-casin digest at 20° to 25°C for the remaining half of the specimens. The minimum incubation time shall be ten days. Sampling shall be as follows:

Medium	Paper Spore Strip	Inoc. Prod. or Inoc. Simulated Prod.
Soybean-casein digest	S-3*	S-3*
Fluid thioglycollate or Alternative thioglycollate		S-3*

* Use a minimum of 20 specimens. Acceptance number is zero.

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Inoculated product or simulated product. For each type of sterilization, the organism used shall be as listed below:

Type of Sterilization	Organism
Steam	B. stearothermophilus
Dry heat	B. stearothermophilus
	B. subtilis
Ethylene oxide	B. subtilis
Gamma radiation	B. pumilus

The number of viable spores shall be sufficient to provide assurance that the product has been sterilized.

Alternate to sterility test. Those sterility processes for this device approved by the FDA shall be acceptable as an alternate to the aforementioned sterility test.

Certificates of biocompatibility. A certificate of compliance, stipulating that offered items meet the biocompatibility test, shall be acceptable, provided that the contractor has supplied acceptable items to the Government within the past three years and that there have been no changes to material formulation and method of processing since actual testing was performed. The certificate shall include the date the testing was performed, the test method and results, the name and title of the person evaluating the results, the current contract number and NSN, a statement certifying that material processing and formulation have not changed since the time that biocompatibility was performed and the signature and title of the person making the certification.

Heavy metals extractable test. The heavy metals extractable test shall be in accordance with one of the following heavy metals tests. The controller shall be tested for heavy metals in accordance with the Physico-Chemical Test of the USP.

Alternate metals extractable test. A 2-gram composite sample shall be obtained from the rubber and plastic components. The material shall be cut into small slivers, so that no thickness exceeds 3mm, and eluted for 72 hours at 50°C, 24 hours at 70°C, 60 minutes at 121°C in one menstruum of 0.9% sodium chloride. Each gram of the material shall be covered with 5 milliliters (ml) of the menstruum. The menstrua are collected, agitated after elution and then submitted to spectographic absorption, spectrophotometric or specific wet analysis for lead, barium, and cadmium. The metals are reported in parts per million. Thus 1.0 ppm in elute is equivalent to extracting 1.0 mg from 200 g of test material dispersed in 1,000 ml of elution menstruum.

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Alternate heavy metals extractable test. Accurately weigh a 5-gram sample for each of the metals (barium, cadmium or lead) into a 1-oz. Vycor crucible. Reduce to ash in a muffle furnace at 650°C for eight hours. Let cool to room temperature and add 15 ml of 1:1 hydrochloric acid. Bring to a boil on a hot plate, allow to boil to 2 to 3 minutes, remove, let cool, and filter through Whatman #2 filter paper into a 50-ml volumetric flask. Rinse the residue on the filter with deionised water and dilute to 50 ml. Prepare standard solutions of 1, 2 and 5 ppm for each of the metals, namely barium, cadmium, and lead. Determine the absorbances of these solutions at appropriate wave lengths using a Hollow Cathode Ba, Cd, or Pb lamp for each of the respective metals and 10X scale expansion on a spectrophotometer, such as the Perkin-Elmer Model 303, Atomic Absorption Spectrophotometer. Determine the absorbance of the sample solutions using the same conditions as with the standards.

Calculation:

$$\frac{A \times F \times \text{dilution}}{W} = \text{ppm barium, cadium, or lead}$$

A = Absorbance

$$F = \text{Factor} = \frac{(\text{Concentration standard in ppm Ba, Cd, or PB})^*}{\text{absorbance}}$$

W = Sample weight in grams.

* The three standards are averaged to give ppm/unit A.

Certificates of heavy metals testing. A certificate of compliance, stipulating that offered items meet the heavy metals extractable testing, shall be acceptable, provided that the contractor has supplied acceptable items to the Government within the past three years and that there have been no changes to material formulation and method of processing since actual testing was performed. The certificate shall include the date the testing was performed, the test method and results, the name and title of the person evaluating the results, the current contract number and NSN, a statement certifying that material processing and formulation have not changed since the time that heavy metals testing was performed and the signature and title of the person making the certification.

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Flow regulator test. The intravenous injection controller shall be tested at flow rates of 5, 25, 50, 100, 150 and 250 ml/hr. Connect an appropriate intravenous set to an intravenous container of 5% Dextrose solution and prime the set. Connect the primed intravenous set to the controller. Fully open the intravenous set control clamp and turn the injection controller dial to open and allow fluid to expel air from the tubing. Turn the dial to off and attach a 20 gauge intravenous set or needle to the controller. Adjust the head height of the container to approximately 30 inches above the needle. Turn the dial until the desired infusion rate is achieved by observing the drip rate in the intravenous set chamber. Allow fluid to flow for at least 10 minutes or until 2 ml are collected, whichever is greater. Repeat the test 3 times at each setting If the calculated flow rate varies by more than 10 percent, the controller shall be rejected.

Instructions. An instruction sheet detailing the proper use of the intravenous injection controller shall be provided with each device.

Workmanship. The intravenous injection controller shall be free from defects which detract from its appearance or impair its serviceability.

PRODUCT QUALITY LEVEL

Visual and dimensional examination. Shall be examined to verify that the design, construction and workmanship are in accordance with applicable requirements.

Materials. The supplier shall be responsible for performing tests to assure that materials are in accordance with requirements specified.

Unit. Package (PG). One package containing forty-eight individually packaged sterile, intravenous injection controllers, as specified, constitutes one unit. The individual package shall be a sealed, peel open type container capable of maintaining sterility of contents unless package is damaged or opened.

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Quality Assurance Provisions.

Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

Records. Records of examinations and tests performed by or for th contractor shall be maintained by the contractor and made available t the Government upon the Government's request, at any time, or from time to time, during the performance of the contract and for a period of three years after delivery of the supplies to which such records relate.

Inspection. Inspection, as used herein, is defined as both examination (such as visual or auditory investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties of the item.

Tests. Tests shall be conducted to determine compliance with specification requirements. Where feasible, the same sample shall be used for the determination of two or more test characteristics.

Sampling for tests. Sampling for tests shall be conducted in accordance with MIL-STD-105 and as specified herein. AQL shall be 1. percent defective. Inspection level shall be S-2. The unit of product shall be one intravenous injection controller.

Contractor certification. The contractor shall certify that the product offered meets the salient characteristics of this description and conforms to the producers' own drawings, specifications, standards, and quality assurance practices. The Government reserves the right to require proof of such conformance prior to first deliver and thereafter as may be otherwise provided for under the provisions of the contract.

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Regulatory requirements.

Federal Food, Drug and Cosmetic Act. If the product covered by this document has been determined by the U.S. Food and Drug Administration to be under its jurisdiction, the offeror/contractor shall comply, and be responsible for compliance by its subcontractors/suppliers, with the requirements of the Federal Food, Drug and Cosmetic Act, as amended, and regulations promulgated thereunder. In addition, the offeror/contractor shall comply, and be responsible for compliance by its subcontractors/suppliers, with the requirements of all other applicable Federal, State, and local statutes, ordinances, and regulations.

Recovered materials. The offeror/contractor is encouraged to use recovered material in accordance with Public Law 94-580 to the maximum extent practical.

Preservation, packaging, packing, labeling, and marking. Unless otherwise specified, preservation, packaging, and packing shall be to a degree of protection to preclude damage to containers and/or contents thereof under normal shipping conditions, handling, etc., involving shipment from the supply source to the receiving activity, plus reshipment from receiving activity, and shall conform to applicable carrier's rules and regulations. Intermediate and exterior package quantities and labeling and marking shall be as specified in the contract and/or order.

NOTE: The following National Stock Number is covered by this document:

NSN	ITEM IDENTIFICATION
6515-01-174-7381	CONTROLLER, DROP RATE, INTRAVENOUS, Disposable, 48s

Ordering data. (Intermediate/exterior package quantities and labeling and marking must be specified in the contract and/or order).

MILITARY INTERESTS:

Custodians:

Army - MD
Navy - MS
Air Force - 03

Preparing Activity:

DoD-MB

Agent:

DLA-DM

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CIVIL AGENCY COORDINATING ACTIVITIES:

VA-OSS
PHS
FDA-MPQAS

Project No. 6515-4258

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